Habtamu Asefa

Software Engineer | Applied AI

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github.com/habasefa · habasefa.github.io/portfolio · LinkedIn

Software engineer with four years of experience across mobile, backend, and applied AI. Built PrepX's mobile app and tutoring agent; adapting speech models for Amharic at Liqawnt Labs.

EXPERIENCE

PrepX | Lead Software Engineer 2024-present

- Built and launched PrepX's React Native/Expo mobile app; students answer approximately **30,000 questions per week**, with approximately **200 students joining each week**.
- Built a retrieval-augmented tutoring agent with NestJS and Mastra, using Qdrant and FastEmbed to retrieve textbook and past-exam material with source references.
- Implemented offline content storage with SQLite, streamed AI responses, and live competition flows; integrated Typesense search across **1,824 curriculum topics**.
- Led a four-person engineering team; contributed to FastAPI backend services and the dashboard.

Liqawnt Labs | AI Research Engineer 2026-present

Low-resource audio startup focused on tokenization, text-to-speech (TTS), and automatic speech recognition (ASR).

- Fine-tuning Pocket TTS on **200 hours of Amharic speech**; preparing audio and text for model adaptation and evaluation.
- Built an internal Amharic evaluation benchmark using word error rate (WER) and native-speaker ratings; ran reinforcement learning (RL) experiments on intonation, prosody, and accent fidelity.
- Implemented VITS encoder-side fine-tuning in PyTorch with monotonic alignment search and a frozen waveform decoder.

Software Engineer | Client Projects 2022-present

Delivered software development and application maintenance across Dymnd, a US bar-exam preparation platform, and Keyring. Implemented timezone-sensitive habit tracking and authentication fixes for Dymnd; resolved critical bugs, upgraded dependencies, and updated features in the bar-exam mobile app.

SELECTED ENGINEERING PROJECTS

GoNumbers | Telecom application

- Developed React Native telecom features with FreeSWITCH/SIP.js calling, WebRTC audio, virtual-number selection, and native call controls.

react-native-ajora | Author & Maintainer 2025-present

- Authored and published a React Native AI chat library used in PrepX, supporting AG-UI streaming, tool-call rendering, and client-side tool execution.

TECHNICAL SKILLS

Languages: Python, TypeScript / JavaScript, Rust, SQL

Backend & UI: FastAPI, Node.js / NestJS, React, React Native

ML & retrieval: PyTorch, Hugging Face Transformers, Qdrant, Typesense

Real-time systems: SIP.js, WebRTC, FreeSWITCH, Socket.IO

Data & deployment: PostgreSQL, SQLite, Linux, Docker, AWS (EC2, S3), GitHub Actions

EDUCATION

Addis Ababa University / AAiT 2024

BSc in Biomedical Engineering. Thesis: UlcerGuard wearable pressure-monitoring prototype.

Connected Arduino foot-pressure sensors to React Native over Bluetooth; streamed readings through a Socket.IO backend to a clinician dashboard, with patient logs and pressure-imbalance alerts.